Cognition 202 (2020) 104333

Contents lists available at ScienceDirect

Cognition

journal homepage: www.elsevier.com/locate/cognit

Evaluating the independence of age, sex, and race in judgment of faces☆

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ARTICLE INFO

Keywords:
Face recognition
Person construal
Garner task

ABSTRACT

Extracting the dimensions of age, sex, race from faces is fundamental for many aspects of social cognition such as person construal, impression formation, and social interaction. While cognitive researchers consider these dimensions to be independent in processing, social psychology researchers have recently demonstrated the emergence of strong interactive patterns between these categories, especially, when social biases are involved. The current study harnessed the classic *Garner's speeded classification task* (Garner, 1974) and *Stroop* task (1935) to evaluate the level of independence between age, sex, and race in a systematic and exhaustive fashion, with an eye on the potential influence of social biases. The degree of separability was evaluated in a pairwise fashion, with each experiment testing one pair. In Experiment 1a and 1b, sex and race were tested with strong (Experiment 1b) or weak (Experiment 1a) social bias. Experiment 2 was set to assess the separability of sex and race. And Experiment 3 was aimed at evaluating the separability of age and sex. The results revealed that neither of the pairs of dimensions produced Garner interferences (and are therefore *separable dimensions*). However, when strong social bias was present, the dimensions did produce redundancy gains and Stroop-like effects, which reflected the presence of abiding social biases. A *holistic-to-analytic* model is presented to account for these results, according to which, the dimensions are initially processed as integral dimensions, but then become gradually separable.

1. Introduction

Imagine that you are introduced to a potential spouse in a speed-dating session. It is likely that your first impression of this person will be formed by three primary dimensions: his or her facial age, sex, and race[1] (Young & Burton, 2017). But how do you extract these attributes? It is widely believed that the perception of social dimensions is held in a categorical manner (Allport, 1954). Categorical perception means that the continuous sensory data impinging on your senses are transformed into distinct abstract labels such as 'Female', or 'Young'. These labels represent well-structured constructs that allow you to go beyond the given sensory datum. In our speed dating example, these dimensions assist you in forming your first impression of the potential mate. Many studies have shown that the perception of age, sex, and race is categorical, rapid and automatic (Bodenhausen & Macrae, 1998; Brewer, 1988; Bruce & Young, 2012; Fitousi, 2017b; Ito & Urland, 2003; Kunda & Spencer, 2003; Kurzban, Tooby, & Cosmides, 2001; Macrae & Bodenhausen, 2000; Martin et al., 2015; Neuberg, 1990); that it precedes the computation of facial identity (Cloutier, Mason, & Macrae, 2005; Cornell, 1974), and that it can emerge spontaneously in a neural network designed to compute identity (Kramer, Young, Day, & Burton, 2017).

A fundamental question in the area of face recognition concerns the psychological processes that govern the extraction of age, sex, and race. Are these dimensions processed independently of each other, or alternatively, do they interact in perception (Freeman, Johnson, Adams Jr, & Ambady, 2012; Quinn & Macrae, 2005) and memory (Fitousi, 2017b; Ito & Urland, 2003)? And if so, how do social biases and stereotypes shape these interactions? These questions are paramount for understanding the formation of attitudes, stereotypes, prejudices and other social phenomena (Freeman & Ambady, 2011; Greenwald, McGhee, & Schwartz, 1998). They are also vital to understanding basic phenomena in face recognition (Bruce & Young, 2012; Calder, Young, Keane, & Dean, 2000). The current study focuses on the dimensions of age, race, and sex. These dimensions have recently started attracting more attention from face researchers (Freeman & Ambady, 2011; Johnson, Freeman, & Pauker, 2012; Martin et al., 2015; Wiese, Kloth, Güllmar, Reichenbach, & Schweinberger, 2012; Wiese & Schweinberger, 2018).

☆ All data sets can be downloaded from https://data.mendeley.com/datasets/rgbh3ttfzn/1.

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[1] Race as a biological concept has been widely criticized, and sex has also been recognized as a complex construct, especially its relation to gender. However, voluminous research has demonstrated the salience of sex and race as social categories for human perceivers, and the ease with which faces can be categorized along these dimensions (Bruce & Young, 2012; Kramer et al., 2017).

https://doi.org/10.1016/j.cognition.2020.104333
Received 15 December 2019; Received in revised form 17 April 2020; Accepted 18 May 2020
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The goal of the current study is to assess the level of their *perceptual independence*.[2]

Face-researchers from cognitive psychology quarters (Bruce & Young, 1986) have argued for the independence of social dimensions. To their way of reckoning, these categories rely on distinctive perceptual information and therefore lack any functional or perceptual overlap (Hill, Bruce, & Akamatsu, 1995; Le Gal & Bruce, 2002). Thus, when observers perceive faces that vary on multiple social categories, relevant and irrelevant categories should not interfere with- or facilitate each other. However, recent studies from social psychology quarters (Freeman et al., 2012; Johnson et al., 2012) have demonstrated the emergence of interaction between the categories of age, sex, and race in face recognition tasks. These studies suggest that social categories can facilitate or interfere with each other (Freeman et al., 2012) due to a myriad of processes from binding in working-memory (Fitousi, 2017b; Martin et al., 2015), to the influence of top-down stereotypes and biases (Freeman et al., 2012), to phenotypic dependencies between dimensions (Johnson et al., 2012). For example, Johnson et al. (2012) argued that: "race and sex are psychologically and phenotypically confounded, affecting social categorization and their efficiency … these findings challenge the notion that social categories are perceived independent of one another …" (p. 116). In the next section, I elaborate on the two seemingly opposed notions of face recognition.

2. The cognitive psychology perspective

The cognitive approach to face recognition has been guided by a classic model of Bruce and Young (1986). This model emphasizes a sharp distinction between two independent processing routes: one route is dedicated to facial identity, and responsible for the retrieval of semantic long-term, abstract representations of familiar faces (FRUs); the second route is dedicated to the processing of facial expression and speech-related movements of the mouth (see Fitousi & Wenger, 2013, for review). The model predicts functional independence between pairs of dimensions that are processed along these two routes (Young, 1998; but see Ganel & Goshen-Gottstein, 2004; Fitousi & Wenger, 2013; Schweinberger & Soukup, 1998; Schweinberger, Burton, & Kelly, 1999). As for the age, sex, and race, these dimensions are considered by Bruce and Young as quintessential semantic dimensions, that are derived directly from the pictorial code and are separate from the identity related semantic information. In their initial presentation of the model, Bruce and Young argued that: "…future studies may allow the separation of 'visually derived semantic codes' into distinct types, produced by different routes. The classification of the sex and approximate age of a face, for example, may involve different processes from those involved in judging that a face appears honest, or resembles that of a particular relative." (p., 109, in Young, 1998). In latter applications of the model, the authors have postulated the independence of social categories. For example, Le Gal & Bruce (2002) argued that sex is processed along the identity route and therefore should be perceived independently of facial expression, a prediction they have confirmed by putting these two dimensions to the Garner test. Moreover, Bruce and her colleagues (Hill et al., 1995) argued that categorization by race is largely influenced by shape information whereas classification by sex is largely based on colour information.

Another important model by Haxby, Hoffman, and Gobbini (2000, 2002) proposed a distributed network with a core system that branches into two routes, one is responsible for the processing of *invariant* aspects of faces (e.g., identity), the second is responsible for *variant* aspects of faces (e.g., expression, lip movements). The former is mainly supported by the *lateral fusiform gyrus*, while the latter is governed mainly by the *superior temporal sulcus* (*STS*). The core system is augmented by an extended system that engages regions of complementing cognitive functions (e.g., attention, emotion). Thus, the interpretation of facial attributes may be influenced by signals from other areas which are not face-specific. This model predicts the interaction of facial dimensions when both are *variant*, or when both are *invariant*, but not when one is variant and the other is not. For example, expression and gaze direction should interact in perception because both are variant. The distributed model focuses on the social dimensions of identity, expression, lip-movements, and eye gaze, and is mute with respect to the dimensions of age, sex, and race. Although one may infer that because these dimensions are invariant dimensions they should interact. However, although the model ascribes a role to prior social knowledge via the extended system, it is too vague to predict how stereotypes, prejudices, and other social knowledge affects the perception of faces. In general, both Bruce and Young and the Haxby et al. models do not address broader aspects of social cognition (e.g., stereotypes), expectations or biases that may affect processing. This naturally since they have been developed to explain face perception at a basic cognitive level.

Recent computational work by Dahl, Rasch, Bülthoff, and Chen (2016), and by Kramer et al. (2017) has provided another valuable perspective on the issue of independence of facial dimensions. These researchers have trained neural networks to identify a small number of facial identities. One unexpected by-product of this training has been the emergence of categorizing ability on sex and race. The emergence of sex and race categorization abilities in a network without an explicit training on these dimensions, and its ability to generalize to untrained faces is remarkable. It suggests the existence of covariations between facial identity features and those of sex and race. This may support integral representation of facial identity and sex, in contrast to the predictions from the Bruce and Young (1986) model. But it also may imply a relatively independent representations of sex and race. If these attributes were also correlated, they would not have emerged as two separate dimensions in the neural networks.[3]

These general claims for independence among the dimensions of age, sex, and race in the cognitive field have been recently contrasted in a series of studies by Weise, Schweinberger and colleagues (Kloth, Damm, Schweinberger, & Wiese, 2015; Schweinberger et al., 2010; Wiese, 2012; Wiese et al., 2012; Wiese & Schweinberger, 2018). These studies documented interesting perceptual interactions that hitherto have not been known. Some of these findings guide the current investigation.

3. The social psychology perspective

The social psychology approach is best represented by the *dynamic interactive model* (Freeman & Ambady, 2011). According to this model, the initial presentation of a face launches simultaneous activations of several competing categories. Along the accrual of evidence, the pattern

[2] The concept of independence has an intuitive appeal. But it is certainly a nontrivial scientific construct (Algom, 2016; Algom, Fitousi, & Eidels, 2017; Algom & Fitousi, 2016; Ashby & Townsend, 1986; Garner & Morton, 1969; Fitousi, 2013; Fitousi & Wenger, 2013). In their seminal paper, Garner and Morton (1969) have argued that *perceptual independence* may not be a unitary concept, but rather, a nomenclature referring to a host of theoretical and operational definitions (see also Ashby & Townsend, 1986; Fitousi & Wenger, 2013). Hence, different tasks and measures may probe different types of independence. These in turn may capture different processing levels and internal representations (Algom & Fitousi, 2016; Miller, 1982). This intuition has guided many of the powerful methodologies advanced in cognitive psychology over the last fifty years or so, including that of the Garner paradigm itself (Garner, 1974), the general recognition theory (GRT, Ashby & Townsend, 1986), and the system factorial technology (SFT, Townsend & Nozawa, 1995). It is one of the central goals of the current study to assess various measures of independence and relate them to different stages of perception and memory.

[3] In the models of Dahl et al. (2016), and Kramer et al. (2017), the independence of dimensions is a by-product of identification. This is different from the dual route model of Bruce & Young.

of activation gradually sharpens into clear interpretation (e.g., old woman), while other alternatives are inhabited. Hence, a confluence of perceptual (bottom-up) and cognitive-social (top-down) factors can generate various types of interactions among social facial dimensions (e.g., age and sex). This model can account for a large body of research which documented interferences between social categories (see Freeman et al., 2012; Freeman & Ambady, 2011). One type of evidence comes from a mouse tracking paradigm in which the moment-to-moment response trends can be examined. This methodology can uncover the influence of irrelevant facial dimensions on those momentary responses. For example, Cloutier, Freeman, and Ambady (2014) showed that the categorizations of age in adult faces, as measured by the trajectory of a mouse toward a label on the screen, were influenced by the irrelevant sex of the face (i.e., male, female), but that the opposite effect was not found.

Support in the dynamic interactive model of Freeman and Ambady (2011) has been provided by Johnson et al. (2012), who demonstrated strong dependencies between the dimensions of sex and race. In their study, categorization of sex was either facilitated or inhibited when the race of the face (e.g., black) shared or did not share either phenotypic or stereotypic information with the certain level of sex (e.g., male). This study documents the influence of both bottom-up perceptual aspects (e.g., covariate pictorial cues between sex and race), as well as top-down stereotypic influences (e.g., pre-experimental attitudes) on the level of dependency between these dimensions.

Another line of research on the processing of multiple social categories comes from sequential response paradigm (Fitousi, 2017b; Martin et al., 2015). In these studies, participants are presented with faces that vary on multiple social dimensions. Participants are asked to focus on one relevant dimension and categorize the faces along this dimension. These studies show that although participants are asked to selectively attend to a relevant dimension, they update the value of the irrelevant dimension and integrate it with the relevant dimensional value across trials. So, for example, if faces vary on both sex and race, and participants are asked to judge the sex of faces, they are faster and less error prone if the specific levels of sex and race completely repeated (Male + Asian → Male + Asian) or completely alternated (Male + Asian → Female Caucasia) from the previous trial, in comparison to cases where only one of the dimension repeated and the other alternated (Male + Asian → Male + Caucasian). These *partial repetition costs* (Hommel, 1998) entail that participants have encoded the value of the irrelevant dimension in the previous trial, and paid attention to it status with respect to the relevant dimension value. The effect also implies the binding of social dimensions in working memory into a temporary episodic representation called face file (see for Fitousi, 2017a, 2017b). A similar mechanism has been demonstrated by Ito and Urland (2003) in an ERP study. The P300, a late ERP component was found to be sensitive to the status of the relevant and irrelevant levels of the social category in the preceding trial.

4. Garner's speeded classification paradigm

Given the almost diametrically opposed conclusions coming from the cognitive and social psychology realms, the question becomes even more pressing: are social dimensions dependent or independent in processing? And if they are, how social biases and stereotypes shape their interaction? The current study offers a systematic assessment of these issues using a powerful methodology — *Garner's speeded classification task* (Garner, 1974; for review see Algom & Fitousi, 2016). The Garner paradigm has been used extensively in the face literature to assess the independence (or its lack thereof) of many pairs of facial dimensions (for review see Algom & Fitousi, 2016). Among the pairs of facial dimensions that have been put to the Garner test are: identity and expression (Fitousi & Wenger, 2013; Ganel & Goshen-Gottstein, 2004; Schweinberger et al., 1999; Schweinberger & Soukup, 1998), sex and age (Quinn & Macrae, 2005), sex and expression (Le Gal & Bruce, 2000), eye gaze and emotion (Graham & LaBar, 2007), and sex and emotion (Atkinson, Tipples, Burt, & Young, 2005). Surprisingly, the dimensional relations between sex, age, and race themselves have been scarcely tested in the Garner paradigm, and the few studies who did deploy the Garner methodology have done this in an isolated and incomplete fashion. Here, I provide, what I believe to be, the first systematic and exhaustive assessment of the perceptual separability of age, sex, and race.

The logic underlying the application of the Garner paradigm is general to any pair of dimensions. Consider the application to the dimensions of race and sex in Experiment 1b, which can also demonstrate how the Garner paradigm can probe the influence of an implicit social bias (Greenwald et al., 1998). In three types of blocks, the observer decided, while timed, whether the face's sex was male or female. In the *Baseline* block, the task-irrelevant race was held at a constant value (e.g., Asian). In the *Filtering* block, the task-irrelevant face's race changed from trial-to-trial in a random fashion (e.g., Asian or Black).

The task for the participant was the same in all the three blocks of trials: to classify the sex of the face as speedily and accurately as possible. The difference in RT between the Baseline and the Filtering blocks defines the measure known as *Garner interference* (Pomerantz, 1983):

$$\text{Garner Interference} = \text{MRT (Filtering)} - \text{MRT (Baseline)} \quad (1)$$

where MRT is the mean latency to classify the sex of the face (as male or female). An analogue formula exists for error. The presence of Garner interference — poorer performance in Filtering than in Baseline — shows that irrelevant variation in race took a toll on performance with the sex of the face, compromising fully selective attention to the sex.

The complete Garner paradigm includes a third type of condition, *Correlation*. In this condition, the irrelevant dimension also changes from trial to trial. However, under Correlation the task-irrelevant dimension varies in a corresponding manner with the task-related dimension. To continue our example, the values of the task-irrelevant race change in tandem with the values of the target task-relevant sex. In one correlated block, male faces were always Black and females were always Asian. In another correlated block, males were always Asian, and females were always Black. Note that in the Garnerian terminology (Algom & Fitousi, 2016; Algom, Dekel, & Pansky, 1996; Fitousi & Algom, 2020; Melara & Algom, 2003; Melara & Mounts, 1993), the former block is called a *positively correlated* (C+) because the faces in this block carry the exact combinational structure that represent a semantic/lexical congruity, namely that Black males are more masculine than Asian males, and that Asian females are more feminine than Black females (Johnson et al., 2012); the latter block (C−) is called a *negatively correlated*, because faces in this block have the opposite intersections of values to that of the bias. It is often expected that if selective attention to the criterial dimension fails, then participants would notice this semantic congruency or incongruency between the dimensions and will exhibit faster processing in the positively correlated block than in the negatively correlated block (see for example, Algom et al., 1996; Fitousi & Algom, 2020). In experiments where no social bias is expected, the two correlated blocks are collapsed into a single correlated block (Algom & Fitousi, 2016).

In general, when the dimensions are not semantically related, and there is only a single correlation block per task, it is often found that performance under Correlation is better than that in Baseline, the difference called, *redundancy gain*. This is because the nominally irrelevant dimension of race now becomes predictive of the relevant dimension of sex. Note also that the presence of redundancy gain also signals the failure of selective attention to the target sex dimension. Now, when a semantic congruity or incongruity exists, and the correlated block is divided into a positively and negatively correlated blocks, the effects are computed slightly different. Performance in the positively correlated block is expected to be faster than baseline – a *redundancy gain*. This effect entails that participants reaped gain from the semantic congruency between the two dimensions; performance in the negatively

correlated block should be worse than baseline, an effect dubbed *redundancy loss*. This effect entails that participants suffered loss due to the semantic mismatch between the dimensions. Both redundancy losses, and redundancy gains entail the failure of selective attention to the relevant dimension. In addition, a pure measure of semantic congruity is computed as a difference in performance between the negatively and positively correlated blocks (Algom et al., 1996; Fitousi & Algom, 2020; Melara & Marks, 1990).

5. Integral and separable dimensions

Garner has identified two patterns of dimensional relations. Pairs of dimensions that produce a Garner interference and redundancy gains (or losses) are called *integral dimensions*. These dimensions are represented as an integrated whole. With this type of dimensions, observers find it difficult to attend to either of the dimensions selectively. Exemplary pairs of integral dimensions are hue and saturation (Garner & Felfoldy, 1970; Hyman & Well, 1967), height and width of rectangles (Weintraub, 1971). In contrast, pairs of dimensions that do not produce Garner interference nor redundancy gains (or losses) are dubbed *separable dimensions*. With this type of dimensions, observers can command perfect selective attention to either of the dimensions. Exemplary pairs of separable dimensions are circle size and the tilt of its diameter (Shepard, 1964), and hue and shape (Garner, 1974). Therefore, the following RT equalities define separable dimensions:

$$\mathrm{MRT(correlation)} = \mathrm{MRT(baseline)} = \mathrm{MRT(filtering)} \quad (2)$$

where MRT is the mean reaction time to respond to the relevant dimension. By contrast, integral dimensions are defined by the following RT inequalities:

$$\mathrm{MRT(correlation)} < \mathrm{MRT(baseline)} < \mathrm{MRT(filtering)} \quad (3)$$

where MRT is again the mean reaction time to respond to the target dimension. In Fig. 1, I illustrate the Garner paradigm along with the two general patterns of results (Fig. 1).

It should be noted that integrality can be an asymmetric property, such that dimension A can interfere with dimension D, but not vice versa (see for example, Algom & Fitousi, 2016; Fitous & Algom, 2020). Also note, Garner and his colleagues have provided converging evidence for the notion of separable versus integral dimensions. For example, in a similarity scaling task, integral and separable dimensions are subjected to different combination rules (Garner, 1974), with the former combined according to Euclidian metric and the latter according to a city-block metric (see Algom & Fitousi, 2016).

6. Stroop-effects in the Garner paradigm

When semantic relations between dimensional values are present, as in the case of colour words and ink colour (RED in blue), the Garner paradigm affords researchers to test for Stroop effects (Algom et al., 1996; Fitousi, 2016; Melara & Algom, 2003; Melara & Mounts, 1993; Van Leeuwen & Lachmann, 2004). This feature is particularly helpful in the current study, where the presence of social biases can generate Stroop-like relations. The Stroop effects can be measured in two ways: (a) as an across-blocks effect, by comparing performance in the positively and negatively correlated blocks, and (b) as a within-block effect, by partitioning the stimuli in the filtering block into congruent and incongruent trials, according to their semantic value, and comparing performance between these two type of trials. Separable dimensions often do not produce Stroop-effects (nor Garner effects), whereas integral dimensions do yield Stroop-effects (as well as Garner effects) (Algom & Fitousi, 2016). The common view has been that Stroop effects cannot occur without Garner effects because a Stroop-effect requires that observers pay attention to the semantic values of the dimensions, and therefore should also notice irrelevant variation (i.e., Garner interference) (Melara & Mounts, 1993). But this maxim has not be fully supported, as some studies recorded Stroop-effects without Garner effects (Van Leeuwen & Bakker, 1995).

7. Experiments

Three experiments have been conducted in the current work. All experiments applied the Garner and Stroop methodologies to the facial dimensions of age, sex, and race. The main goal was to provide an exhaustive assessment of their dimensional relations, with an eye on the potential influence of a social bias on the level of separability. In addition to the Garner measures, Stroop-like measures have also been applied to gauge the presence or absence of semantic conflict as a marker of social bias. The degree of separability was evaluated in a pairwise fashion. Each experiment was designed to test one pair. In Experiment 1a and 1b, sex and race were put into the Garner test. The dimensional separability of these social categories was investigated. It was assumed that the presence of a strong social bias (Experiment 1b) should increase the likelihood of integrality of these dimensions. Experiment 2 was set to assess the separability of sex and race. And Experiment 3 was aimed at evaluating the separability of age and sex.

8. Experiment 1a and 1b

8.1. Sex and race

Experiments 1a and 1b were designed to assess the separability of sex and race with the presence of strong (Experiment 1b) or relatively weak (Experiment 1a) social bias. Cognitive models of face recognition (Bruce & Young, 1986) predict these social categories to be independent in processing. Hill et al. (1995) showed that the computation of race is largely based on shape information, whereas the computation of sex is based on colour information. This strengthen the likelihood of their independence, because colour and shape themselves are separable dimensions (Garner, 1974). In contrast, social psychology models such as the *dynamic interactive model* of Freeman and Ambady (2011) predict that sex and race should appear as integral dimensions due to the presence of psychological and phenotypic confounds (Johnson et al., 2012). Is it the case, that sex and race appear as dependent dimensions when researchers intersect dimensional levels that create a strong social bias, but as independent dimensions when the levels give rise to weak or no social bias?

Here, for the first time, the two dimensions are pitted one against the other in the Garner speeded classification paradigm. This is done under two different contexts, one in which a social bias is strongly present, and one in which it is relatively weak. To manipulate the degree of social bias, I capitalized on recent findings of Johnson et al. (2011), who showed that the degree of interaction between sex and race depend on the choice of intersected values. These researchers showed that judgments of sex were differentially affected by race. For males, classification RTs increased as the race changed from Black to Caucasian to Asian; while for females, classification RTs decreased as a function of these changes in race. These results suggested that Asian females are perceived as more feminine than black females and black males are perceived as more masculine than Asian males, while Caucasian males and females are perceived somewhat in between. In Experiment 1a, the dimensional levels selected for race were 'Caucasian' and 'Asian', whereas in Experiment 1b the levels were Asian and Black (see Appendix A). The intersection of these dimensional values with the levels of sex (male, female) should give rise to a weak potential cognitive bias in Experiment 1a, and a strong one in Experiment 1b. The two experiments can potentially uncover the possible influence of stereotypic influence on the level of dimensional integrality of the two dimensions.

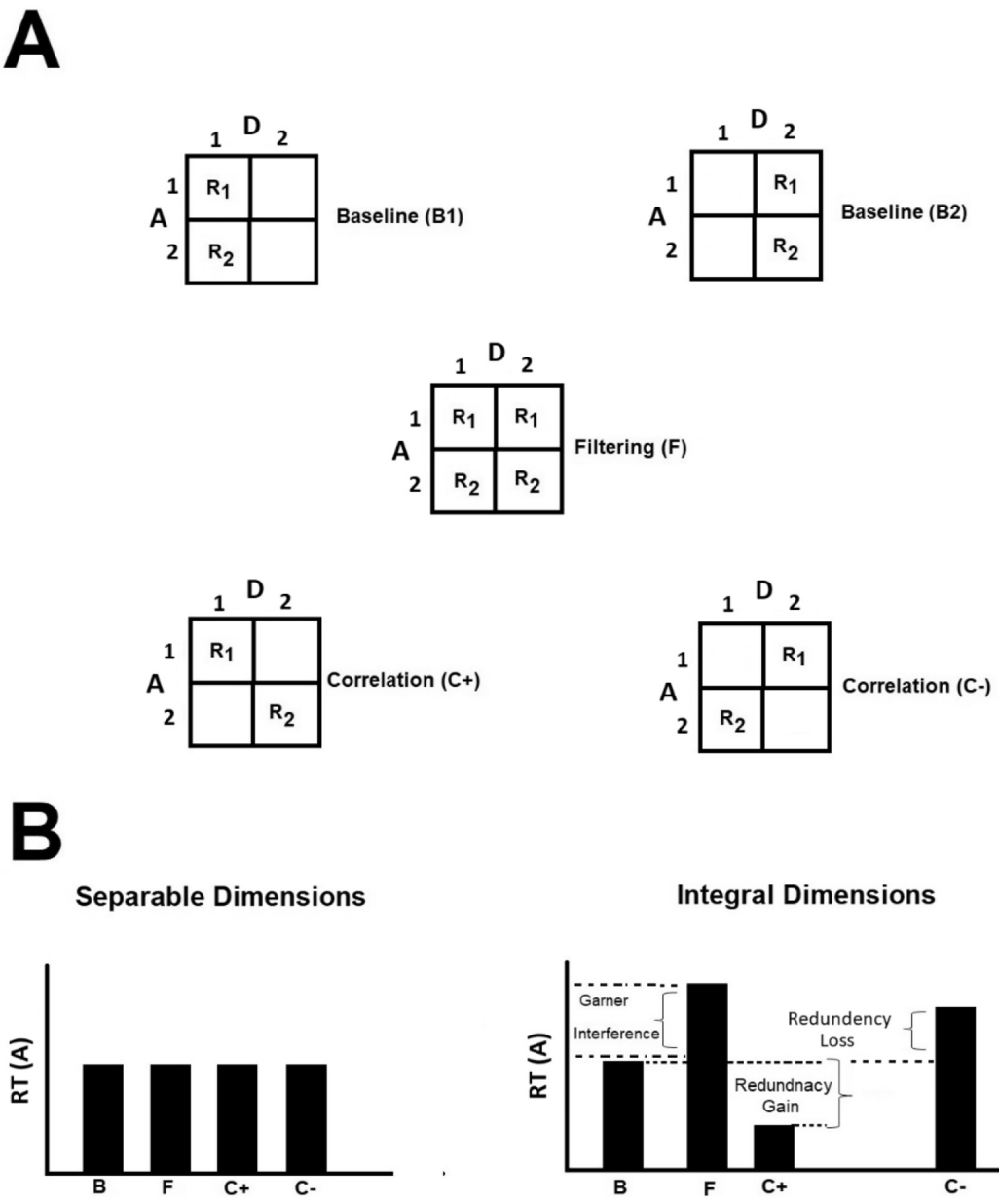


Fig. 1. Schematics of Garner's speeded classification paradigm used in the study. The four basic conditions — baseline (B), filtering (F), positive correlation (C+) and negative correlation (C−) — are depicted on the top. R1 and R2 depict the correct responses to the two ranges of values of the relevant dimension. Two prototypical outcomes, tapping integral and separable processing, are shown at the bottom.

8.2. Methods

8.2.1. Participants

A total of sixty young students from Ariel University took part in the Experiment. 30 in Experiment 1a (14 men and 16 women mean age = 26.3, SD = 3.1) and 30 in Experiment 1b (13 men and 17 women, mean age = 26.2, SD = 2.4). All had normal or corrected-to-normal vision and perfect use of their two hands. All gave their signed consent.

8.2.2. Stimuli

The face images were taken from a well-known image archive (Ma, Correll, & Wittenbrink, 2015). The free GIMP software was used to standardize all the faces by cropping the hair of each face and placing it within a standard oval shape of approximately 7 cm × 9 cm. Viewed from an approximate distance of 56 cm the stimuli subtended 9.09° × 7.12° visual angle. The faces varied on sex (male, female) and race (Asian vs. Caucasian in Experiment 1a, and Asian versus Black in Experiment 1b). The expression of all faces was neutral. Levels of brightness, size, and shape were kept constant across all faces. There were 80 different faces in total, 40 in Experiment 1a, and 40 in Experiment 1b (see Appendix A). Theses faces were evenly distributed across the four possible intersections of sex and race (i.e., 10 faces per intersection).

8.2.3. Procedure and design

Participant sat in front of a laptop computer in a dimly lit room. On each trial a single face was randomly chosen by the computer and presented at the center of the screen until response. It was then replaced by a blank screen for 300 ms and then a new face was presented. Each participant engaged in two tasks: classification of race, and classification of sex. The order of the tasks was random. Each task consisted of 5 blocks (two baselines, two correlations, and one filtering). Baseline or correlated blocks included 20 trials, and filtering block included 40

trials. A complete cycle of all 5 Garner blocks consisted of 120 trials for a given task. Each task cycle was performed three times in a random order. Overall, each participant completed 720 trials. Participants responded with either their right or left hand by pressing the button keys ("M" or "Z").

8.2.3.1. Categorization by race. Participants classified the faces along the race dimension. In the filtering block, race and sex varied orthogonally and all combinations of race (Asian vs. Caucasian in Experiment 1a, and Black vs. Caucasian in Experiment 1b) and sex (male, female) were presented randomly. In the baselines blocks sex varied and race was held fix. There were two baseline blocks: in one baseline the irrelevant dimension (sex) was restricted to male faces, and in the second baseline it was restricted to female faces. In the correlated blocks, race covaried with sex. There were two correlated blocks. These blocks reflected either positive or negative correlations of the dimensional values depending on the social bias they conveyed. In Experiment 1a, a *positively correlated block* (C+), was one in which only Asian-female and Caucasian-male faces were presented. A *negatively correlated block* (C−), was one in which only Asian-male and Caucasian-female faces were presented. In Experiment 1b, a *positively correlated block* (C+), was one in which only Asian-Female and Black-Male faces are presented. A *negatively correlated block* (C−), was one in which only Asian-Male and Black-Female faces are presented.

8.2.3.2. Categorization by sex. Participants classified the faces along the sex dimension. In the filtering block, sex and race varied orthogonally, meaning that all combinations of sex and race were presented in a random fashion, such that all combinations of race (Asian vs. Caucasian in Experiment 1a, and Black vs. Caucasian, in Experiment 1b) and sex (male, female) were presented randomly. In the baseline blocks, sex varied but race was held fixed in one value. There were two baseline blocks: in one block, the irrelevant dimension (race) was held fixed in one level (e.g., Asian), and in the second baseline block it was held fixed in the other level, depending on the levels of race used in Experiment 1a, and 1b. The structure of the correlated blocks was identical to that used in judgments of race.

Participants were asked to respond as fast and as accurate as they could. Each experimental block was separated from the preceding block by a short break of 20 s.

8.3. Results

8.3.1. Experiment 1a

All data sets for all the experiments in this work can be downloaded from https://data.mendeley.com/datasets/rgbh3ttfzn/1.

Trials in which RTs were longer than 2800 ms or shorter than 150 ms were removed from the analysis (3.6%). Incorrect trials were also removed (11.25%). Mean reaction times (RTs) were computed for each participant in each condition. Table 1 presents mean RTs and error rates in the experiment. All Bayesian analysis were performed using the *BayesFactor R package* (Morey & Rouder, 2018). Bayes factors (BF) are reported as evidence for the best fitting model relative to an alternative. BF > 3 or smaller than 1/3 represents substantial evidence; BF between 1/3 and 3 represents only weak evidence (Jeffreys, 1939/1961; see also Dienes, 2014). All ANOVA models include an individual subject main factor (Morey, Rouder, Jamil, & Morey, 2015). The Bayesian statistics employed here, in contrast to the traditional one, can inform us about the null (Dienes, 2014). This is extremely important when assessing separability, which is operationalized as a null effect.

Table 1
Experiment 1a, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in the baseline, filtering, and correlation conditions. SE = standard error.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Baseline	881	41	12.1	1.5	876	42	10.7	2.2
Filtering	916	44	12.5	1.9	874	42	10.9	2.4
Correlation +	870	44	11.1	1.3	823	44	4.5	0.7
Correlation −	922	44	11.1	3.3	842	42	15.6	4.2

8.3.1.1. Garner interferences. Garner interferences were assessed by subjecting mean RTs to a two-way Bayesian ANOVA with Task (sex, race) and Block (baseline, filtering) as factors. Data from the two baseline blocks were collapsed. All the Bayesian ANOVA models pointed to the absence of an effect ($0.01 < BFs < 0.22$), and therefore provided strong positive evidence for the lack of a Garner interference. Evidence also supported the hypothesis that the two dimensions were of equal relative discriminability, as the overall speed of the two tasks was comparable (BF = 4.5). A similar analysis on error rate revealed comparable results, with all Bayesian ANOVA models showing evidence in support for the absence of Garner interference ($0.012 < BFs < 0.24$).

8.3.1.2. Redundancy gains and losses. In the *positive correlation block* the levels of sex and race of faces created congruent combinations (Asian Females, and Caucasian Males), whereas in the *negative correlation block* the levels create incongruent combinations (Caucasian Females and Asian Males) according to the social bias reported by Johnson et al. (2012). To test for redundancy gains, a two-way Bayesian ANOVA with Task (sex, race) and Block (positive correlation, baseline) was performed. The results supported the absence of any effect ($0.016 < BFs < 0.25$). A comparable analysis on accuracy revealed substantial evidence for a model with Block, Task, and Block × Task effects (BF = 24,635). A set of *Bayesian t-test* revealed redundancy gains (i.e., lower error rates in the positively correlated blocks than in baseline) for judgments of race (BF = 3.03), but not for judgments of sex (BF = 0.22).

To test for redundancy losses a Bayesian ANOVA with Task (sex, race) and Block (negative correlation, baseline) was conducted on RTs. The evidence favoured the absence of any effect ($0.016 < BFs < 0.30$), as did the analyses on error rates ($0.008 < BFs < 0.17$).

Another way in which the Garner paradigm can uncover a social bias is through a Stroop-like congruency effect in the correlated blocks. Such an effect may document improved performance in a positively congruent block relative to a negatively correlated block (Algom et al., 1996; Fitousi, 2016). A Bayesian ANOVA was conducted with Task (race, sex) and Block (positively correlated, negatively correlated). The results provided evidence for no across-block Stroop effect ($0.09 < BFs < 0.90$). A similar analysis on error rates pointed to similar conclusions ($0.49 < BFs < 2.02$).

8.3.1.3. Stroop effects. These effects are based on performance within the filtering blocks. Mean RTs and error rates were computed for each of the four types of faces in the filtering blocks (see Table 2). The four stimuli were then mapped to congruent (Asian Female, Caucasian Male) and incongruent (Caucasian Female, Asian Male) stimuli according to the stereotypic structure reported by Johnson et al. (2012). Stroop-like effects were assessed by entering the mean RTs into a two-way Bayesian ANOVA with Task (sex, race) and Congruency (congruent, incongruent) as factors. Although a Stroop effect was numerically present (see Table 3), the statistical analyses provided strong evidence against the emergence of Stroop-like effects ($0.01 < BFs < 0.27$). A similar result was obtained in the error rate ($0.1 < BFs < 0.59$).

Table 2
Experiment 1a, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in filtering condition. SE = standard error.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Asian male	963	54	10.9	3.3	870	60	15.0	4.1
Caucasian male	842	44	2.7	1.1	907	46	7.1	2.1
Asian female	980	47	22.4	2.4	827	39	5.4	1.4
Caucasian female	921	49	14.1	3.5	909	49	16.0	4.5

Table 3
Experiment 1a, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in Stroop congruent and incongruent trials. SE = standard error.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Congruent	900	44	12.5	1.4	864	40	6.3	1.4
Incongruent	941	46	12.5	3.3	896	49	15.5	4.2
Stroop effect	+41		0	+32		+9.2		

8.4. Results

8.4.1. Experiment 1b

Trials in which RTs were longer than 2800 ms or shorter than 150 ms were removed from the analysis (3.9%). Incorrect trials were also removed (14.8%). Mean reaction times (RTs) and error rates were computed for each participant in each condition. They are presented in Table 4.

8.4.1.1. Garner interferences. Garner interferences were assessed by subjecting mean RTs to a two-way Bayesian ANOVA with Task (sex, race) and Block (baseline, filtering) as factors. Data from the two baseline blocks were collapsed (Algom et al., 1996; Fitousi & Wenger, 2013). The evidence for a model with Garner interference – a difference in performance between filtering and baseline blocks – was very weak (BF = 0.19). A model with Task as main effect (BF = 137) exhibited superior evidence to all other models, implying faster categorization of faces according to race (mean = 692 ms) than according to sex (mean = 808 ms). Comparable analyses on error rates yielded weak evidence for any effect ($0.05 < BFs < 0.24$).

8.4.1.2. Redundancy gains and redundancy losses. In the *positive correlation block* the levels of sex and race of faces created congruent combinations (Black Males, and Asian Females), whereas in the *negative correlation block* the levels formed incongruent combinations (Black Females, Asian Males). According to Johnson et al. (2012) these dimensional intersections can elicit a strong stereotype. A two-way Bayesian ANOVA with Task (sex, race) and Block (positive correlation, baseline) was performed. Models with Task (BF = 54), Block + Task (BF = 47) and Block + Task + Block × Task (BF = 13.43) effects were strongly supported. Most importantly, evidence for a redundancy gain (better performance in positive correlation blocks than in baseline) was evident in both the race (BF = 16.8) and sex (BF = 6.5) categorization tasks. Categorization by race was faster than by sex (BF = 3,986,477). A two-way Bayesian ANOVA with Task (sex, race) and Block (negative correlation, baseline) supported models with Task (BF = 58.21), Block + Task (BF = 17.67) and Block + Task + Block × Task (BF = 5.07) effects. Most importantly, a redundancy loss (worse performance in negative correlation blocks than in baseline) was evident in categorization of sex (BF = 4.61), but not in categorization of race (BF = 0.56). Redundancy loss and redundancy gain effects imply the emergence of a semantic analysis of the facial categories. Participants reaped gain when the stimuli varied according to a social bias but suffered loss when the stimuli varied against it.

Another way by which the Garner paradigm can probe a social bias is through a Stroop-like congruency effect in the correlated blocks. Such an effect may document faster performance in a positively correlated block than in negatively correlated block (Algom et al., 1996; Fitousi, 2016). A Bayesian ANOVA was conducted with Task (race, sex) and Block (positively correlated, negatively correlated). The results showed that a model with a Task × Block received the greatest support (BF = 508). Congruity effects were evident in both categorization of race (BF = 546) and sex (BF = 187). These effects entail the emergence of a social bias in perceiving race and sex. Participants were faster to perceive a combination of social features that constitute a social bias (e.g., Black Male, Asian Female) than a combination that does not (Asian Male, Black Female).

8.4.1.3. Stroop effects. Mean RTs were computed for each of the four types of faces in the filtering blocks (see Table 5). The four stimuli were then mapped to congruent (Asian Female, Black Male) and incongruent (Asian Male, Black Female) stimuli according to the stereotypic structure reported by Johnson et al. (2012). Stroop-like effects were assessed by entering the mean RTs into a two-way Bayesian ANOVA with Task (sex, race) and Congruency (congruent, incongruent) as factors. The strongest model was one with a main effect of Task (BF = 466), next was a model with Task and Congruency as main effects (BF = 311), and then a model with Task, Congruency and their interaction (BF = 95). A *Bayesian t-test* confirmed that classifications by race were slower than classifications by sex (BF = 9,605,941,188). Table 6 gives mean RTs and error rates in congruent and incongruent trials. Most importantly, substantial Stroop-like congruency effects emerged in the classifications by race (BF = 3.89) and by sex (BF = 2179), supporting the earlier finding of a Stroop-like effect in the correlation blocks, and implying the activation of the expected stereotype in both filtering and correlated tasks. A comparable analysis on error rate did not reveal any effect.

Table 4
Experiment 1b, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in the baseline, filtering, and correlation conditions.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SD	M	SD	M	SE	M	SD
Baseline	799	32	16.3	1.7	697	27	12.6	2.0
Filtering	817	28	18.0	1.7	687	31	13.2	1.8
Correlation +	749	32	12.3	2.3	638	24	11.5	2.5
Correlation −	844	38	16.6	1.7	718	26	13.4	2.6

Table 5
Experiment 1b, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in filtering blocks. SE = standard error.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Asian male	853	41	9.2	1.7	677	29	9.6	2.1
Asian female	823	26	20.3	3.0	653	29	10.9	2.1
Black male	758	29	10.6	2.2	689	36	16.9	2.2
Black female	869	31	32.0	3.1	736	39	16.3	2.1

Table 6
Experiment 1b, mean RTs (in ms) and proportions of error (%) for judgments of race and sex in Stroop congruent and incongruent trials. SE = standard error.

	Sex				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Congruent	785	26	15.4	2.4	671	31	13.5	2.1
Incongruent	854	32	20.6	1.8	705	32	12.9	1.9
Stroop effect	+69**		0		+34**		−0.6	

** BF > 3

8.5. Discussion

Experiments 1a and 1b were set to test the perceptual separability of sex and race in Garner's speeded classification task (Garner, 1974) under two separate contexts – with strong (Experiment 1b) or weak (Experiment 1a) social bias. The results from the two experiments documented the absence of a Garner interference under the two contexts, such that classification of race was not interfered by irrelevant variation on sex. And similarly, categorizations of sex were not disrupted by irrelevant variation on race. Participants could pay perfect selective attention to the criterial dimension and were immune to interference from the irrelevant dimension. This held true whether a strong or weak social bias has been introduced into the stimuli set. By this pattern the dimensions are best described as *separable dimensions*.

However, the correlated blocks and the Stroop measures in the filtering blocks told a somewhat more complicated but perfectly sensible story. When social bias was weak (Experiment 1a) the correlation blocks also indicated strict separability, because performance in these blocks was comparable to that in baseline and filtering, and no Stroop-like effects (either in the correlated blocks or in the filtering blocks) have been recorded. However, when strong social bias was present (Experiment 1b) participants: (a) reaped redundancy gains when the dimensional values varied according to those expected by the social bias (in the positively correlated blocks) relative to a baseline, (b) suffered redundancy losses, when the dimensional values varied against the expected social bias (in the negatively correlated blocks), (c) enjoyed an improved performance in blocks in which the dimensional values of faces were congruent with the social bias (positively correlated) compared to blocks in which they were incongruent with the social bias (negatively correlated), and (d) exhibited substantial Stroop effects in the filtering blocks. These four effects were symmetric, surfacing in both the race and sex categorization tasks.

The results point to two main outcomes. First, sex and race do not yield Garner interferences, and this holds true irrespective of the strength of social bias. These dimensions can be therefore dubbed *separable dimensions*. Second, these dimensions do produce Stroop and redundancy gains under the context of strong social bias (Experiment 1b). How can these results be reconciled? First, the absence of Garner and Stroop effects with weak social bias strongly supports the separability of these dimensions in this case. But how can one explain the results under strong social bias? It is likely that strong social bias dramatically alters the processing of these dimensions. In this case, there are two alternative accounts, both refer to the processing dynamics of the dimensions across time (Fitousi, 2019; Van Leeuwen & Bakker, 1995).

According an *analytic-to-holistic* explanation, the dimensions are separable from the outset, and become holistic or integral only at later stages, where social bias effects influence behaviour. According to an alternative *holistic-to-analytic* account, the dimensions are integral from the outset, as reflected by the presence of a social bias, but then become progressively separable. There is strong evidence to support the latter account with the key finding being the recording of Stroop effects without Garner effects. Although the finding of a Stroop without Garner is quite rare in the literature (Algom & Fitousi, 2016), it is not without precedents (Fitousi, 2016; van Leeuwen & Bakker, 1995). It has been often argued that the emergence of Garner interference is necessary for the emergence of a Stroop effect (Melara & Marks, 1990), simply because in order to notice semantic conflict (Stroop effects) observers first must notice variation (Garner interference). But according to van Leeuwen and Bakker (1995) this logic is not compulsory, if one considers the holistic-to-analytic dynamics of perception (Fitousi, 2019; Kimchi, 1992) by which, in the course of perception, dimensions can start as integral or holistic but then progressively become analytic or separable.

Evidence for such temporal dynamics comes from the observation that Stroop effects occur much earlier than Garner effects. An ERP study (Boenke, Ohl, Nikolaev, Lachmann, & van Leeuwen, 2009) has shown that with geometrical shapes, Garner interference affects processing about 330 ms after stimulus onset but Stroop-congruence effects start earlier, about 170 ms after stimulus onset, with hardly an overlap between them. It is therefore likely that in the first 300 ms of perceptions, dimensions are integral (resulting in Stroop effects), whereas after 300 ms, they become separable (so no Garner interference is recorded). The upshot is that under the context of strong social bias (Experiment 1b), sex and race are perceived holistically at the beginning, generate Stroop and redundancy effects, but then become progressively separable, and yielding no Garner interference. None of these patterns occurs under the context of weak social bias (Experiment 1a). In that case, the dimensions are perceived analytically along the entire processing. All this boils down to the conclusion that stereotypes, and social biases are not functioning as top-down influences, but rather as hardwired tendencies that critically mold early perception, dictating holistic or integral perception of the constituent dimensions. But these influences are weakening with time to allow a relatively independent access to the two dimensions.

The current results replicate the findings of Johnson et al. (2012) who reported that race either facilitated or hindered categorization of sex, depending on the congruency of the social bias. Moreover, Johnson et al. (2012) demonstrated this social bias only for judgments of sex, while the current study extends their validity to judgments of race as well. The current results offer a solution to the puzzle we introduced at the outset, according to which researchers from the cognitive quarters predict race and sex to be independent in processing (Bruce & Young, 1986; Hill et al., 1995) whereas researchers from the social psychology quarters predict their interaction. Who is correct? It appears that both. According to the Garner test, the dimensions are perceptually separable, as they exhibit the absence of Garner interference. However, when the stimuli set contain an implicit social bias, the dimensions can demonstrate a breakdown of independence and selective attention, as reflected in redundancy and Stroop-like effects, stemming from influences of social bias at the early stages of processing. These patterns are best accounted for by a holistic-to-analytic dynamics.

9. Experiment 2

9.1. Age and race

This experiment aimed at assessing the dimensional relations of age and race. To the best of my knowledge, this is the first application of the Garner test to these dimensions. Karnadewi and Lipp (2011) have deployed the Garner paradigm pairing age, race, and sex – in different experimental blocks – with facial emotion. They found that age, race, and sex interfered with the classification of facial emotion, but not vice versa. Thus, it is still not clear whether age and race themselves are integral or separable with each other. The current experiment will allow us to test this issue directly.

9.2. Methods

9.2.1. Participants

A new sample of 30 young students from Ariel University took part in this experiment (16 females and 14 males, Mean age = 24.21 SD = 2.66). All had normal or corrected-to-normal vision and perfect use of their two hands. All gave their signed consent.

9.2.1.1. Stimuli. Stimuli were 40 realistic grayscale images of faces presented over black background (see Appendix B). The face images were taken with permission from a known face images archives (Ma et al., 2015). The faces varied on age (young, old) and race (Black, Caucasian). All faces were female faces with neutral expression. The young faces were 20–30 years old, whereas the old faces were 60–70 years old. All other details are as in Experiment 1a, and 1b.

9.2.1.2. Procedure and design. All details are comparable to those in Experiment 1a and 1b.

9.2.1.2.1. Categorization by race. In the filtering block, race and age varied orthogonally and all combinations of race (Black, Caucasian) and age (young, old) were presented randomly. In the baseline blocks, race varied orthogonally from trial-to-trial, but age was kept at one level (e.g., old). There were two baseline blocks: in the first baseline the irrelevant dimension (age) was restricted to old faces, and in the second baseline the irrelevant dimension was restricted to young faces. In the correlated blocks, race covaried with sex. In one correlated block, only old Caucasian and young black faces were presented. In a second correlated block, only young Caucasian and old Black faces were presented. Since, to the best of my knowledge there is no social stereotype that involves the intersections of these dimensions, the correlated blocks were not assigned to positive or negative correlation but were kept separate to explore possible biases.

9.2.1.2.2. Categorization by age. Participants classified the age of the face and ignore its race. In the filtering block, age and race varied orthogonally, meaning that all combinations of race and age dimensional levels were presented in a random fashion. In the baselines blocks, age varied orthogonally from trial-to-trial, but race was held fixed at one value. There were two baseline blocks: in one block the irrelevant dimension (race) was restricted to Black, and in the second baseline block it was restricted to Caucasian. In the correlated blocks, age covaried with race, with the exact correlated blocks described in the previous section. All the other details are as in Experiment 1a, and 1b.

9.3. Results

The data of two participants were removed from analysis due to high error rate (> 40%). Trials in which RTs are longer than 2800 ms or shorter than 150 ms were removed from analysis (3.4%). Incorrect trials were also removed (12.6%). Mean reaction times (RTs) were computed for each participant in each condition. Table 7 presents the mean RTs and error rates in the two tasks.

9.3.1. Garner interferences

Garner interferences were assessed by subjecting mean RTs to a two-way Bayesian ANOVA with Task (age, race) and Block (baseline, filtering) as factors. Data from the two baseline blocks were collapsed. The evidence for all models was low ($0.014 < BFs < 0.28$), refuting the possibility of the emergence of Garner interference. Strong evidence was provided for the hypothesis that the two dimensions were of equal relative discriminability (BF = 3.84).

9.3.2. Redundancy gains and redundancy losses

Redundancy gains and redundancy losses were assessed in a two-way Bayesian ANOVA with Task (age, race) and Block (baseline, correlated) as factors. I kept the two correlated blocks separate to allow the detection of potential social bias. I tested separately whether each one of them produced redundancy gains or losses. The evidence was against the former ($0.02 < BFs < 0.38$) and the latter ($0.01 < BFs < 0.23$). A comparable analysis on the error rate revealed similar results ($0.07 < BFs < 1.32$) and ($0.01 < BFs < 0.32$) for the two type of blocks, respectively.

Table 7
Experiment 2, mean RTs (in ms) and proportions of error (%) for judgments of race and age in the baseline, filtering, and correlation conditions. SE = standard error. Correlation 1 consists of old Caucasian and Young Black faces. Correlation 2 consists of young Caucasian and Old Black faces.

	Age				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Baseline	806	49	9.1	2.1	777	45	10.1	2.1
Filtering	814	45	8.9	2.2	763	45	9.5	2.3
Correlation 1	799	45	5.9	1.9	718	43	5.1	1.7
Correlation 2	816	50	11.6	3.4	789	46	12.7	2.0

Next, I have tested the presence of a potential social bias, checking whether the two correlation blocks reveal a congruency effect across blocks. A two-way Bayesian ANOVA with Task (age, race) and Block (positive, negative correlation) was performed. The analysis on RT provided evidence against such a possibility ($0.053 < BFs < 0.36$). But the analysis on error rate supported such a difference, providing evidence for a main effect of Block (BF = 6.9). Error rates were larger in the second correlation block than in the first correlation block (see Table 7).

To further test the possibility that a cognitive bias does exist in the perception of age and race, I assessed a congruency effect in the filtering block based on these dimensional values (see Table 8). The stimuli were mapped to congruent (old Caucasian, young Black) and incongruent (young Caucasian, old Black) combinations according to the pattern found in the correlation blocks. Table 9 gives the Means RT and error rates for the congruent and incongruent trials. A two-way Bayesian ANVOA with Task (age, race) and Congruency (congruent, incongruent) provided evidence against Stroop-like effects and therefore refuted the presence of any social bias ($0.02 < BFs < 0.37$). A comparable analysis on error rate led to similar conclusions ($0.04 < BFs < 0.84$) with error rates.

9.4. Discussion

The results of Experiment 2 revealed that the categories of age and race are separable dimensions. Participant could attend selectively to age while ignoring irrelevant variation on race, and vice versa, a conclusion that was corroborated by the absence of a Garner interferences. Participants have not reaped gain from the covariation of the two dimensions. In the correlated blocks it was found that certain combinations (old Caucasian and young Black) were classified more accurately

Table 8
Experiment 2, mean RTs (in ms) and proportions of error (%) for judgments of race and age in the filtering blocks. SE = standard error.

	Age				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Caucasian old	784	44	5.8	1.7	788	50	6.7	2.8
Caucasian young	773	46	9.1	3.9	740	51	7.8	2.8
Black old	845	50	12.8	3.2	769	44	16.8	2.3
Black young	852	49	8.0	2.4	754	43	7.2	2.4

Table 9
Experiment 2, mean RTs (in ms) and proportions of error (%) for judgments of age and race in Stroop congruent and incongruent trials. SE = standard error.

	Age				Race			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Congruent	814	44	6.8	1.8	771	45	6.9	2.5
Incongruent	819	49	10.9	3.1	754	46	12.2	2.3
Stroop effect	+5		+4.1	ns.	−17		+5.3	

than others (young Caucasian and old Black). But the nature of this bias is not clear, and it might be incidental, as the partition of stimuli to these values in the filtering block into congruent and incongruent trials did not reveal any Stroop effect. Overall, these results provide strong evidence for the perceptual separability of age and race.

10. Experiment 3

10.1. Sex and age

Experiment 3 aimed at evaluating the dimensional separability of sex and age. Previous studies have reached inconsistent conclusions regarding the perceptual separability of these dimensions. Deploying the Garner paradigm, Quinn and Macrae (2005) documented an asymmetric interference from age to sex, but not vice versa. In another study, Kloth et al. (2015) predicted that observers would take advantage of the presence or absence of smooth skin texture and use it as a clue to facial age, which in turn, should affect judgments of sex. This prediction was corroborated. It took participants longer to judge the sex of an old female than a young female, while the reverse held true for males. This finding uncovered a perceptual (social?) bias (young female, old male/old female, young male). This type of bias has also been demonstrated by Schweinberger et al. (2010). They have shown that female test faces are judged as younger when preceded by old female/young male adaptor faces as compared to when preceded by young female/old male adaptor faces; whereas male test faces are judged older when preceded by old female/young male adaptor faces as compared to when preceded by young female/old male adaptor faces. The nature of this interaction is still not well understood and can be due to perceptual or social biases. Other types of (non-semantic) interactions have been recorded by Fitousi (2017b) in a binding task and by Cloutier et al. (2014) in a mouse tracking task.

It should be noted that the application of the Garner paradigm by Quinn and Macrae (2005) suffered from several limitations which the current application is designed to obviate: (a) their study did not include the correlation task, which is part and parcel of the Garner paradigm (Algom & Fitousi, 2016), (b) age was a far more discriminable dimension than sex, rendering their conclusion of asymmetric integrality dubious (Algom & Fitousi, 2016; Fitousi, 2016; Melara & Mounts, 1994), (c) social bias and Stroop-like effects were not tested for, and (d) the study employed traditional statistics, and therefore could not provide positive evidence for the absence of Garner interference if existed (Dienes, 2014). The current study overcomes all these shortcomings by adding the correlation block, assessing the potential artefact of relative discriminability, assessing the presence of social bias by testing for Stroop-like congruency effects, and deploying Bayesian statistics to evaluate potential null effects.

10.2. Methods

10.2.1. Participants

A new group of 30 undergraduate students from Ariel University participated (mean age = 27.27, SD = 3.09, females = 16, males = 14). Participants received course credit for their participation. All reported normal or corrected-to-normal vision. All participants filled in an informed consent.

10.2.2. Stimuli

Stimuli were 40 realistic grayscale images of faces presented over black background (see Appendix C). All images were imported from dedicated websites in the public domain (Ma et al., 2015; Minear & Park, 2004). The faces varied according to sex (male vs. female) and age (young vs. old). There was an equal number of faces from each of the four possible intersections of the two categories. The young faces were 20–30 years old, whereas the old faces were 60–70 years old. The facial expression (neutral), race (Caucasian), brightness, and other aspects of the faces were kept constant. The free GIMP software was used to standardize all the faces by cropping the hair of each face and placing it within a standard oval shape as in the previous experiments. All other aspects are identical to those reported in Experiments 1a and 1b.

10.2.3. Procedure and design

All aspects are identical to those reported in Experiments 1a, 1b.

10.2.3.1. Age classifications. In the filtering block age and sex varied orthogonally and all combinations of age (old, young) and sex (male, female) were presented. In the baselines blocks age varied but sex was held fixed. There were two baseline blocks: in the first baseline the irrelevant dimension (sex) was restricted to male faces, and in the second baseline block it was restricted to female faces. In the correlated blocks, age covaried with sex. The assignment of the correlated blocks to positive and negative correlation was made based on the social bias reported by Kloth et al. (2015) and Schweinberger et al. (2010). In the positively correlated block, only old-male and young-woman images were presented. In the negatively correlated block, only images of old-females and young-males were presented.

10.2.3.2. Sex classifications. In the filtering block age and sex varied orthogonally. In the baselines blocks sex varied but age was held fix. There were two baseline blocks: in one block the irrelevant dimension (age) was restricted to old faces, and in the second baseline block, it was restricted to young faces. The correlated blocks were identical to those constructed in the age categorization task, but the task was that of classification by sex.

10.3. Results

Trials in which RTs are longer than 2800 ms or shorter than 150 ms were removed from analysis (1.2%). Incorrect trials were also removed (6.7%). Mean reaction times (RTs) were computed for each participant in each condition. Table 10 presents mean RTs and error rates in each condition and task.

Table 10
Experiment 3, mean RTs (in ms) and proportions of error (%) for judgments of age and sex in the baseline, filtering, and correlation conditions. SE = standard error.

	Age				Sex			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Baseline	697	23	8.5	2.0	825	23	10.1	1.0
Filtering	700	22	8.7	1.8	845	24	10.0	0.8
Correlation +	675	22	10.3	2.9	678	21	5.7	1.3
Correlation −	705	22	4.9	1.3	702	30	6.3	1.2

10.3.1. Garner interferences

Garner interferences were assessed by subjecting mean RTs to a two-way Bayesian ANOVA (Morey et al., 2015) with Task (age, sex) and Block (baseline, filtering) as factors. Data from the two baseline blocks were collapsed (see Fitousi & Wenger, 2013). The model for an effect of Task was superior to all other models (BF = 177,603). It was 4.66 times more probable than a model with two main effects (Task + Block), and 16.34 times more probable than a model with two main effects and an interaction (Task + Block + Task × Block). The most favourable model provided evidence for a Task effect, namely, that age was classified more rapidly than sex (see also Quinn & Macrae, 2005). Most importantly, together with the model of a main effect of block (BF = 0.21), these results provide strong evidence against the emergence of Garner interference. Comparable analyses on accuracy rates yield evidence for no effects (0.14 < BFs < 0.28). In sum, the Bayesian analyses documented higher discriminability of age than sex, and most importantly, positive evidence for the absence of Garner interference in both tasks. The latter strongly supports the separability of age and sex.

10.3.2. Redundancy gains and losses

A two-way Bayesian ANOVA with Task (age, sex) and Block (baseline, correlated) as factors was held. I first examined the redundancy effects with the positively correlated blocks which included faces of old males and young females. The strongest evidence was provided by a model with main effects of Block and Task, and their Block × Task interaction (BF = 1175). The effect of Task pointed again to an overall faster categorization latency of age than sex. A set of Bayesian *t*-test showed that the source of the interaction was a redundancy gain effect in classification of sex (BF = 1,240,385,932), but not in categorization of age (BF = 0.8). A comparable analysis with the negatively correlated blocks, which consisted of faces of old females and young males, pointed to a similar conclusion (BF = 18.04), with a redundancy gain in judgment of sex (BF = 25,906), but not in classification of age (BF = 0.22). This result entails that when judging the sex of the face, participants were faster in the correlated conditions than in baseline, reaping gain from the covariation of the two dimensions, irrespective of the specific combination of dimensional values. This was not the case when participants were judging the age of the face.

The intrusion of age into sex judgments in the correlation blocks is in line with the findings of Kloth et al. (2015). However, this study argued for a perceptual (social?) bias in judgment of sex, according to which faces of old males and young females were classified faster than faces of young males and old females, because skin texture served as a cue to sex in the former, but not in the latter. Although numerically present, these differences were present in mean RTs, the statistical analyses provided strong evidence against it (0.01 < BFs < 0.33).

10.3.3. Stroop-like effects

Mean RTs and error rates for each of the four type of faces in the two tasks are presented on Table 11. I first tested for the cognitive bias reported by Kloth et al. (2015) in judgments of sex. This bias has been confirmed in the current results, as faces of young females were responded to more efficiently than faces of old females, in both RTs (BF = 11.3) and error rates (BF = 5.15); and faces of old males were responded to more efficiently than faces of young males in both RTs (BF = 6.1) and error rates (BF = 6.3). However, the evidence was against such bias in judgments of age (0.24 < BFs < 2.1). The interaction between dimensions can also be viewed as a Stroop effect. I mapped the four type of faces in the filtering blocks into what should be expected to be congruent (old male, young female) and incongruent (young male, old female) stimuli according to the Kloth et al. (2015) reported cognitive bias (see Table 12). I then conducted a two-way Bayesian ANOVA with Task (age, sex) and Congruency (congruent, incongruent) as factors. A model with main effects of Congruency and Task (BF = 952,093) provided the strongest evidence for a congruency effects in both tasks. A set of Bayesian *t*-test further confirmed the presence of congruency effects in judgments of age (BF = 47) and in classification of sex (BF = 58). No such effects were recorded with errors.

Table 11
Experiment 3, mean RTs (in ms) and proportions of error (%) for judgments of age and sex in the filtering blocks. SE = standard error.

	Age				Sex			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Male old	696	25	12.8	2.7	849	27	8.9	1.8
Female old	688	22	7.4	1.4	863	26	11.3	1.9
Male young	752	29	6.0	1.4	903	27	14.9	1.7
Female young	666	26	8.8	3.1	784	27	5.2	1.2

Table 12
Experiment 3, mean RTs (in ms) and proportions of error (%) for judgments of age and sex in Stroop congruent and incongruent trials. SE = standard error.

	Age				Sex			
	RT		Error (%)		RT		Error (%)	
	M	SE	M	SE	M	SE	M	SE
Congruent	679	22	10.8	2.9	815	25	6.9	1.2
Incongruent	720	23	6.7	1.2	880	24	13.0	1.3
Stroop effect	+41**		−4.1		+65**		+6.1	

** BF > 3

10.4. Discussion

The results of Experiment 3 reveal several important outcomes. First, the dimensions of age and sex produced no Garner interferences. Thus, when classifying faces along the sex dimension, participants were not distracted by irrelevant variation on age, exhibiting good selective attention; and similarly, when classifying faces according to age, participants were immune to interference from irrelevant variations on the dimension of sex, demonstrating good selectivity for the criterial dimension. The Bayesian statistics yielded positive evidence for the absence of Garner interferences in both tasks, and therefore positive evidence for their symmetric separability. This result, however, was accompanied by two other findings: (a) the emergence of redundancy gains reaped by participants in judgments of sex (but absent for classification of age), and (b) the presence of Stroop-like effects related to a specific perceptual or social bias, in the filtering blocks of both tasks. The finding of Stroop-like effects corroborated and extended earlier studies (Kloth et al., 2015; Schweinberger et al., 2010) which documented a bias in judgments of sex by which females are associated with young age and males with old age. This bias was recorded here in judgments of either age or sex.

At first look, the results present a conundrum. According to one metric (Garner), the dimensions are independent of each other, but according to another metric (Stroop), the dimensions interact. How can this be reconciled? The key observation here is the finding of Stroop effects (and redundancy gains) without Garner interference. This pattern has already been found in Experiment 1b, where it has been given a satisfactory explanation. The same explanation is viable here. Although rare, Stroop effects without Garner effects have been recorded (Fitousi, 2016; Van Leeuwen & Bakker, 1995). The best account of this pattern is provided by a *holistic-to-analytic* explanation (Van Leeuwen & Bakker, 1995). This account relies on two premises: (a) perception is holistic at the outset, but then becomes progressively analytic (Fitousi, 2019;

Kimchi, 1992), and (b) that Stroop effects occur much earlier than Garner effects (Boenke et al., 2009). According this account, the dimensions of age and sex are perceived holistically at early stages of processing, hence the Stroop effects, but then at later stages, perception becomes gradually analytic, resulting in the absence of Garner interreferences. A comparable pattern has been observed in Experiment 1b, with the dimensions of sex and race, where a strong social bias has also been gauged through Stroop-like effects, and where no Garner interferences were recorded.

Taken as a whole, the results from the three experiments may point to a more general pattern in the processing of social facial categories. When a social bias is present, social dimensions undergo through gradual change in the degree of cross talk across time. From the vantage point of the *two-systems approach* (Kahneman, 2011), such a pattern makes a lot of sense. At the outset, observers need to perceive the dimension of a face holistically in order to form a first impression, or to produce fast and frugal signs of danger. During this time-window the system is vulnerable to cognitive biases, prejudices, and stereotypes. At this stage, participants are fully aware of the semantic and conceptual meanings of the dimensional values involved, but then *system two* takes the rein, and more analytic and bias-free operations take over. At this stage, participants can selectively attend to a required dimension, and prevent distraction from irrelevant dimension.

In a previous study, Quinn and Macrae (2005) found an asymmetric Garner interference from age to sex. The current study did not replicate this asymmetric Garner interference. Based on the Garner and priming methodologies, Quinn and Macrae (2005) concluded that: "perceivers are able to attend to features that support multiple categorizations, yet they show little evidence of having extracted the conceptual meaning of these nonrelevant dimensions." (p. 474). The current results are at odds with this conclusion, pointing to a more intricate story. The crosstalk between the dimensions is not a single concept but amenable to several levels of processing and stages during which participants are aware of the semantic and conceptual meanings of the dimensions. If participants were not aware of these meanings, Stroop-like effects would not have emerged in the current study.

11. General discussion

Extracting the dimensions of age, sex, race from faces is fundamental for many aspects of face cognition (Dahl et al., 2016; Kramer et al., 2017; Fitousi & Wenger, 2013; Freeman & Ambady, 2011; Young & Burton, 2017). These dimensions are perceived at a glance and exert immense influence on downstream processes, such as impression formation, stereotypic thinking, and social biases (Greenwald et al., 1998). An important question in the face recognition literature concerns the level of dependence/independence between facial dimensions (Bruce & Young, 1986; Haxby et al., 2002). The question of independence bears far-reaching consequences for theorizing on face recognition as well as to practical applications. The current study is the first to offer an exhaustive and complete investigation of the dimensional relations between the facial categories of age, sex, and race, using one of the most powerful and popular methodologies – Garner speeded classification task (Garner, 1974; for review see Algom & Fitousi, 2016). This methodology was supplemented here by measures of Stroop effects (Stroop, 1935), which extended our ability to gauge semantic and conceptual levels of processing.

Experiment 1a and 1b, tested the dimensions of race and sex. The dimensions did not produce any Garner interferences whether a strong (Experiment 1b) or weak (Experiment 1a) social bias was present. The dimensions did produce Stroop and redundancy gains only under the context of strong social bias (Experiment 1b). Experiment 2 examined the dimensions of age and race. The dimensions did not produce Garner interferences, redundancy gain, or Stroop-like effects. Experiment 3 investigated the dimensions of age and sex. These dimensions produced no Garner interferences but have given rise to redundancy gains and Stroop-like effects. The absence of Garner interferences in all experiments entails that all three pairs are *separable dimensions*. But the emergence of Stroop-like effects (as markers of social biases) in Experiment 1b and Experiment 3, point to the central role of stereotypes in generating interdimensional crosstalk at the semantic level with the pairs of race and sex, and age and sex. For these pairs, the key observation is the emergence of Stroop effects (and redundancy gains) without Garner effects. This finding is quite rare (Algom & Fitousi, 2016), but not without is precedents (Le Gal & Bruce, 2002; Van Leeuwen & Bakker, 1995). How can this unique pattern be explained?

This pattern is best accounted for by a *holistic-to-analytic* model (Van Leeuwen & Bakker, 1995), which implies that in the course of processing, these facial categories shift from being processed as integral dimensions to separable dimensions. The model assumes that perception is holistic at the beginning, but becomes gradually analytic (Fitousi, 2019; Kimchi, 1992). It also relies on the observation that Stroop effects occur much earlier than Garner effects (Boenke et al., 2009). According to this account, race and sex (Experiment 1b) and age and sex (Experiment 3) are processed holistically at early stages (as evident by the emergence of the Stroop effects); but analytically at later stages, (as signalled by the absence of Garner interreferences).

This explanation is consistent with the *two-systems approach* (Evans & Stanovich, 2013; Kahneman, 2011), which postulates the existence of two processing systems in human cognition, a fast, automatic, and inaccurate system operating early; and a slow, controlled, and rule-based system, operating late. According to this approach, at initial stages, the first system process featural and dimensional aspects of faces holistically, in order to form a global impression, detect potential hazards, and assess in-group or out-group membership. Because these stages are governed by the fast and primitive system, perception is vulnerable to cognitive biases, prejudices, and stereotypes. The emergence of Stroop-like effects likely takes place in these stages where participants are aware of the semantic and conceptual meanings of relevant and irrelevant information. These stages are replaced by more analytic, and bias-free operations, as the second controlled system takes over. At this stage, participants can selectively attend to a required dimension, and prevent distraction from irrelevant information. This account is also consistent with recent brain imaging studies (Amodio, 2014; Quadflieg et al., 2011) showing that stereotypic information modulates early stages of processing involving person perception and conflict resolution.

11.1. Separating perceptual and semantic codes of age, sex, and race

In the classic model of Bruce and Young (1986), the dimensions of age, sex, and race are ascribed to a type of processing code called *visually derived semantic code.* This code is "…readily formed, and can be useful in remembering unfamiliar faces…" (p., 102). Bruce and Young contrasted this code from *identity-specific semantic code*, which is derived from the structural analysis of familiar faces. One insight that emerges from the current study is that age, sex, and race are not only characterized by semantic (categorical) codes, but also by physical codes, and these two can be dissociated. Although age, sex, and race are considered as categorical and therefore semantic dimensions (Freeman & Ambady, 2011), they are initially generated through the computation of specific spatial and pictorial cues in the image (Johnson et al., 2012). And as such, can be handled in a purely bottom-up fashion, without recourse to top-down operations. Sex and race have been shown to surface spontaneously in neural networks trained to recognize a small number of identities (Dahl et al., 2016; Kramer et al., 2017). This is possible because these dimensions are grounded in perceptual non-semantic, image properties. The difficulty of separating perceptual from semantic (categorical) aspects of social dimensions in human responses has been acknowledged by social psychology investigators as a serious obstacle in the understanding of social categorization (Freeman et al., 2012). By demonstrating the specific experimental effects for obtaining

perceptual separability and conceptual integrality, the current study not only supports the distinction, but also shows how they can be dissociated.

This conclusion implies that independence or its violation can occur at the different processing levels or codes. Garner and Morton (1969) have noted that *perceptual independence* is not a unitary concept, but rather, a nomenclature referring to a host of theoretical and operational definitions (see also Ashby & Townsend, 1986; Fitousi & Wenger, 2013). For example, the dimensions of sex and emotion have also been found to be separable dimensions in the Garner paradigm, (Le Gal & Bruce, 2002), but also to elicit interactive Stroop-like patterns among dimensional values, such as men/Angry and female/happy (Becker, Kenrick, Neuberg, Blackwell, & Smith, 2007), or female/surprised and male/angry (Le Gal & Bruce, 2002). The upshot is that different tasks and measures can tap into varieties of independence (see Fitousi & Wenger, 2013), which correspond to several processing levels, internal representations, and processing codes.

11.2. Separating perceptual and working memory processes in social categorization

Interactive patterns among facial dimensions can emerge also due to the involvement of working memory processes (Fitousi, 2017a, 2017b; Ito & Urland, 2003). These are needed to maintain the phenomenal continuity of the face across time and space (Hommel, 1998; Kahneman, Treisman, & Gibbs, 1992). The operation of such mechanisms has been recently demonstrated in a series of studies (Fitousi, 2017a, 2017b; Martin et al., 2015), showing that when participants are judging faces on one social dimension (e.g., sex), they are still sensitive to the variation on the irrelevant social dimension (e.g., race) in the previous trial. Participants are slower and more error prone when only one of the relevant or irrelevant dimensional values alternate from previous to- current trial, in comparison to trials in which both dimensional values change or both alternate together. These *partial repetition costs* (Hommel, 1998) effects have also been found in the current study (see Appendix D).

Fitousi (2017a, 2017b) has recently proposed that such partial repetition costs reflect the binding of facial features into a temporary, episodic representation called *face files* in working memory. The aim of these representations is to maintain the continuity of the face over time and space in memory. Evidence to such an updating process has been adduced by Ito and Urland (2003) in an ERP study. These researchers recorded a late P300 component in the categorization of sex and race which was affected by the preceding stimuli. P300 were larger when the target social category differed from the previous stimuli on the relevant-task dimension. Ito and Urland concluded that “at this point, working memory models were clearly influenced by the social category information to which participants were explicitly attending as well as social category information that was not explicitly task relevant.” (p.621). In addition, Ito and Urland also found specific early markers (N100, P200) of sex and race categorization, but these were not contingent on the dimensional values of the previous face. The presence of working-memory updating of the irrelevant dimension that entails dependence, along with markers of social categorization which are independent of the previous dimension values strengthen the current position that varieties of independence exists in the course of processing facial dimensions.

CRediT authorship contribution statement

Daniel Fitousi: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Visualization, Investigation, Validation, Writing - review & editing.

Appendix A

A sample of 24 out of the 40 faces used in Experiment 1a (top panel) and in Experiment 1b (bottom panel).



Appendix B

A sample of 24 out of the 40 faces used in Experiment 2.



Appendix C

A sample of 24 out of the 40 faces used in Experiment 3.



Appendix D

Analyses of partial repetition effects

Experiment 1a judgments of sex in complete repetition trials (both race and sex repeated together) were responded to faster than *partial repetition* trials (BF = 64, BF = 182); similarly, complete alteration trials were responded faster than partial repetition trials (BF = 2.19, BF = 4.25). In Experiment 2, judgments of age in complete repetition trials (both race and age repeated together) were responded to faster than *partial repetition* trials (BF = 604, BF = 180); similarly, complete alteration trials in which both age and race were responded faster than partial repetition trials (BF = 89.96, BF = 6.15). In Experiment 3, judgments of sex in complete repetition trials (both age and sex repeated together) were responded to faster than *partial repetition* trials (BF = 219, BF = 1034); and complete alteration trials were responded faster than partial repetition trials (BF = 6.5, BF = 13.07).

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